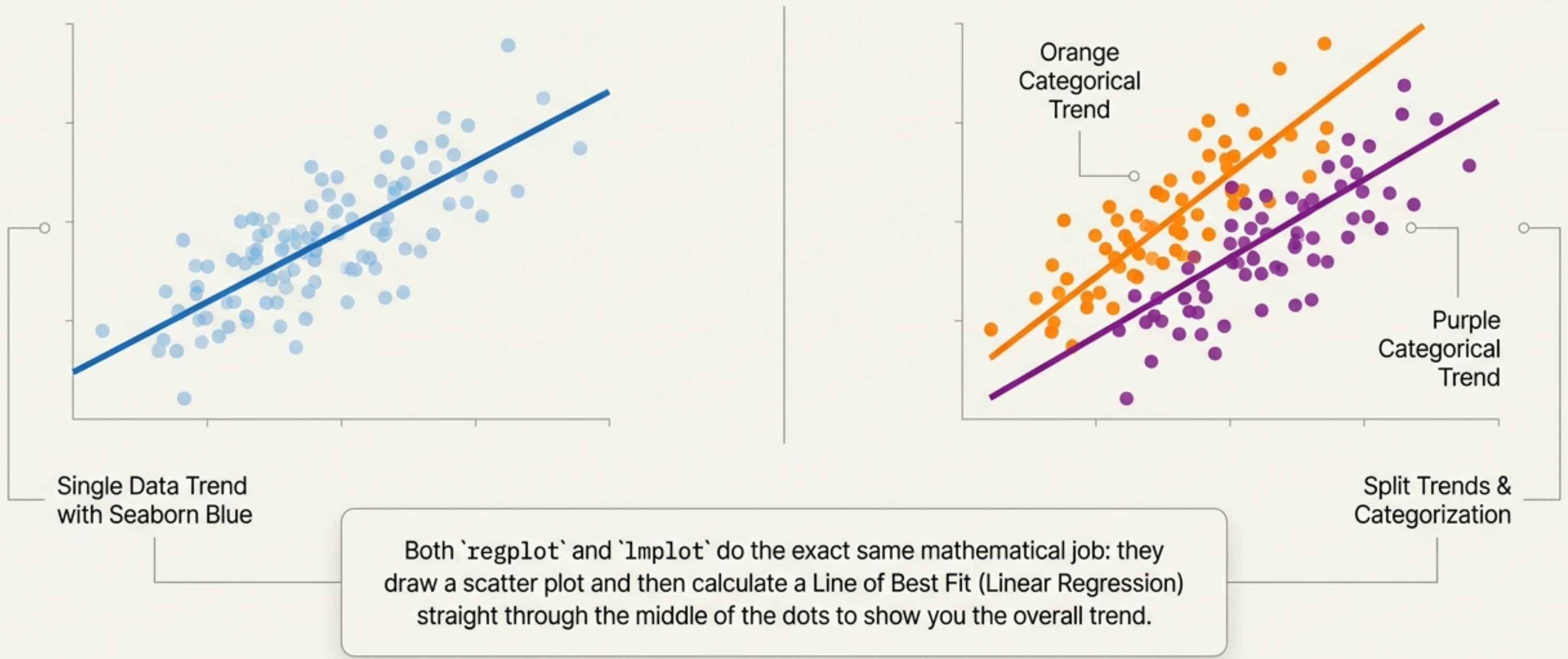
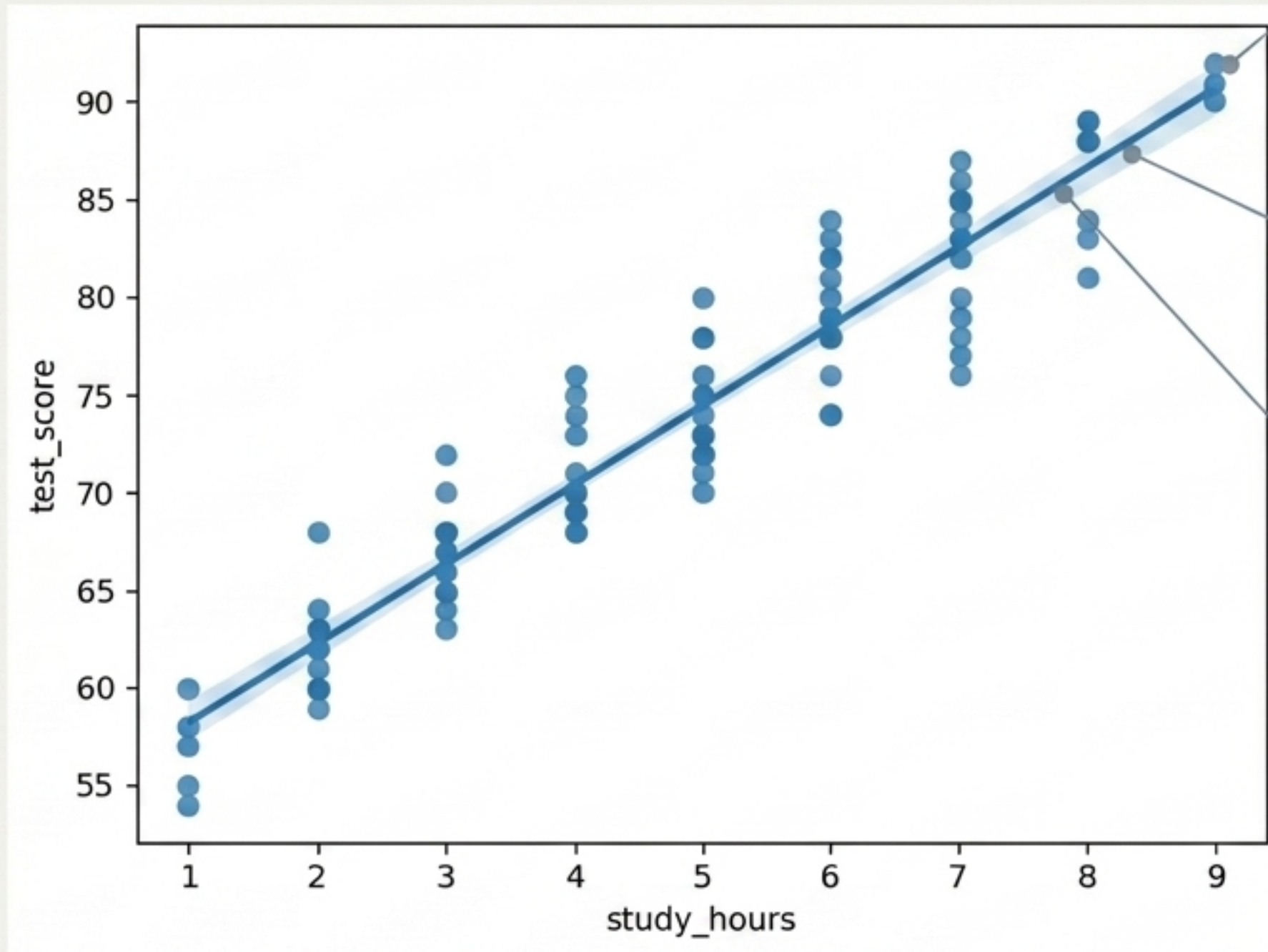


Mastering Regression Plots in Seaborn



The difference between them is purely structural. It is the exact same relationship as `scatterplot` vs `relplot`.

Anatomy of a Seaborn Regression Plot



Scatter Plot

The raw data points plotted on the X and Y axes.

Line of Best Fit

The linear regression calculated straight through the middle of the dots to reveal the overall trend.

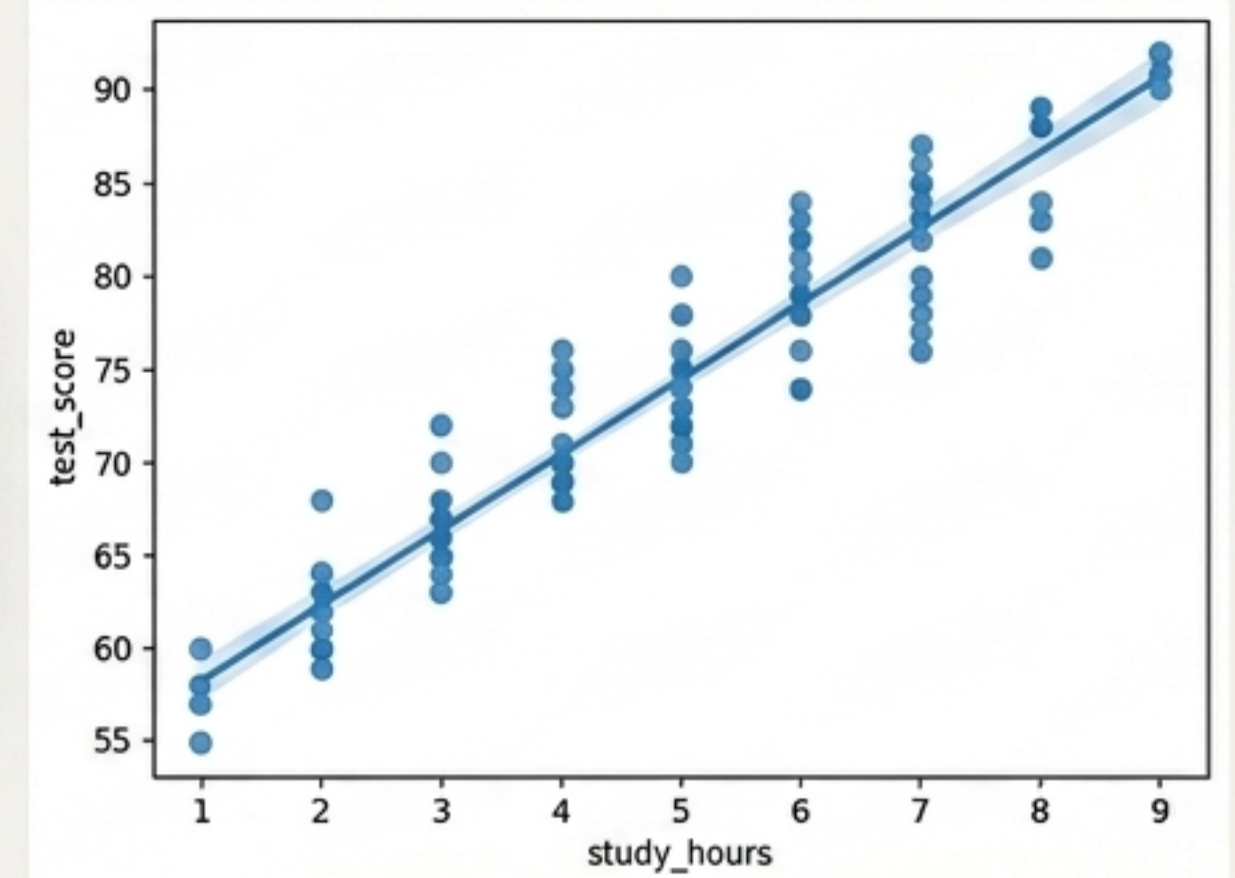
Confidence Interval

The shaded area shows how "sure" Seaborn is about the trend (similar to error bars on a bar chart).

The Basic Building Block: `sns.regplot()`



```
# Shows one overall trend: Does studying increase test scores for everyone?  
sns.regplot(data=student, x="study_hours", y="test_score")  
plt.show()
```



What it is:

Regression Plot

Level:

Axes-Level
function.

How it works:

It draws all your data points
and puts one single line
through them to show the
general trend.

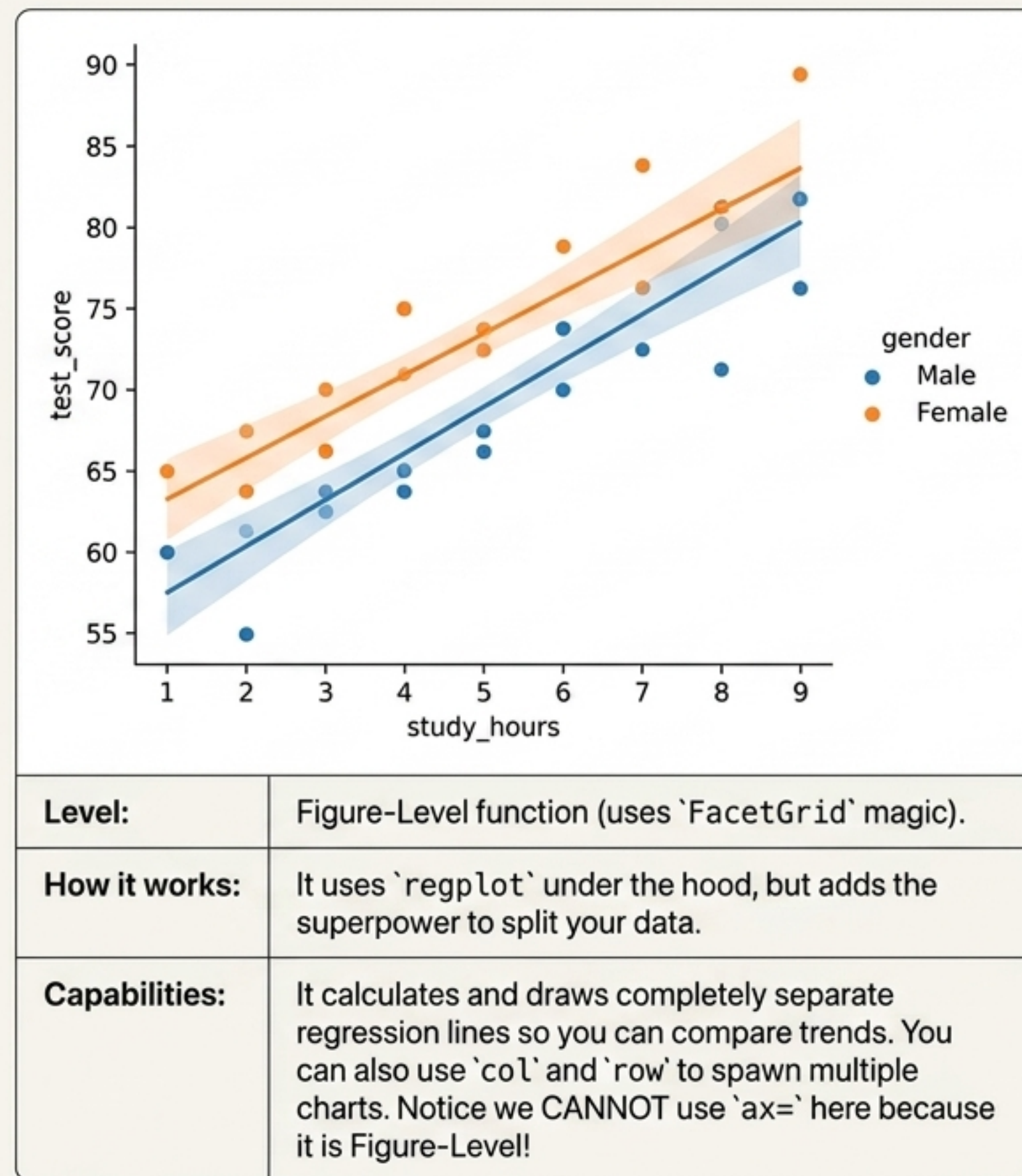
Capabilities:

Because it is axes-level, you can easily put
it inside a specific Matplotlib box using
the `ax=` parameter. It cannot use `col`
or `row` to make sub-plots.

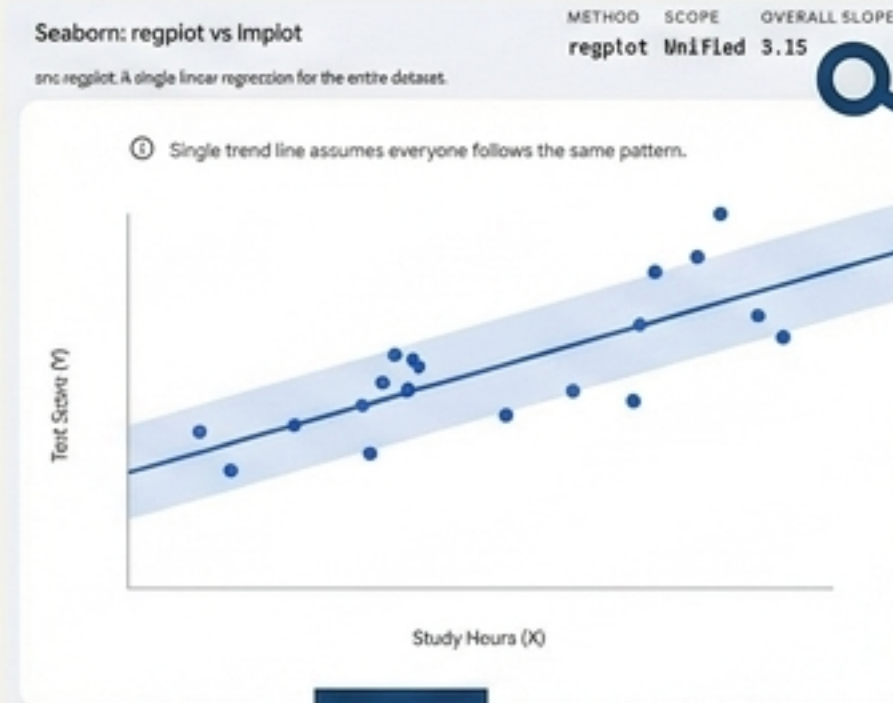
The Powerful Manager: 'sns.lmplot()'

```
# Shows multiple trends: Does studying increase  
scores FASTER for boys or girls?  
sns.lmplot(data=student, x="study_hours",  
           y="test_score", hue="gender")  
plt.show()
```

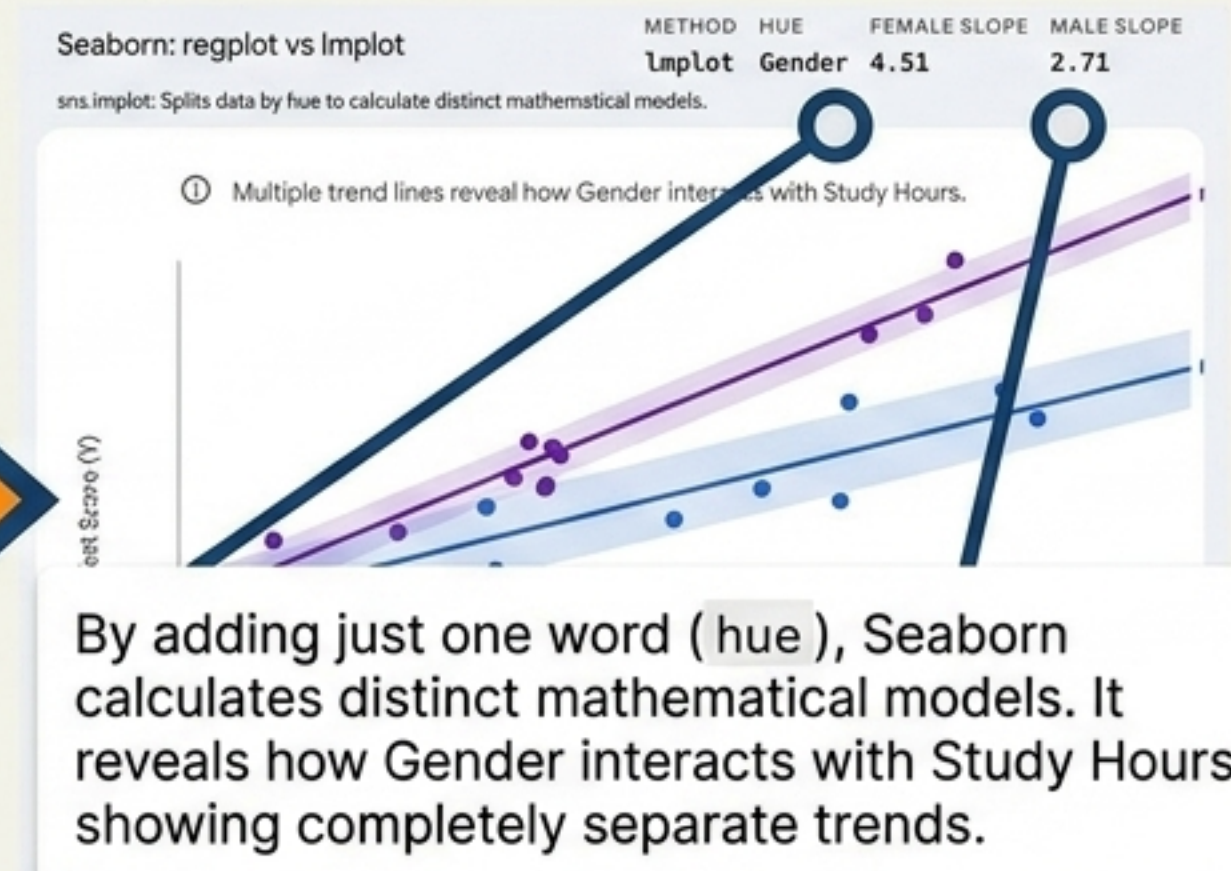
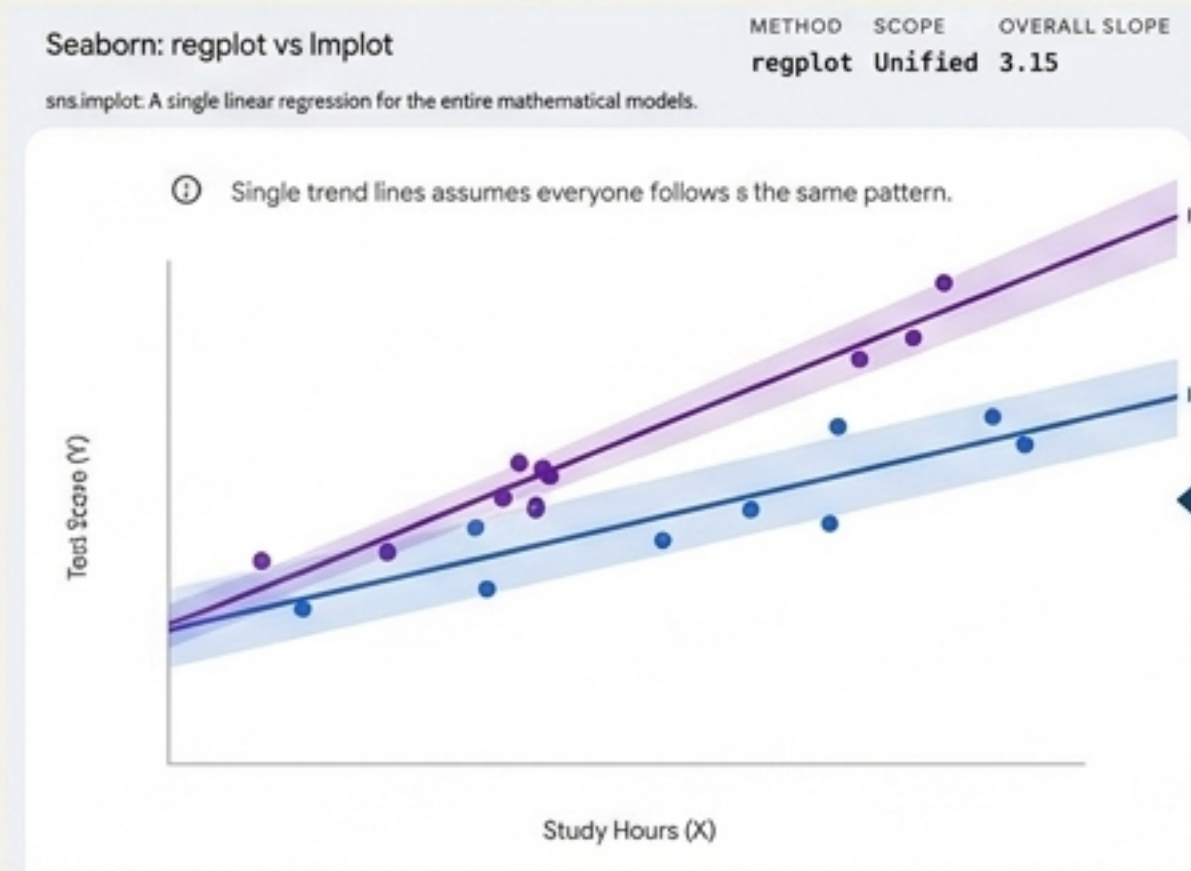
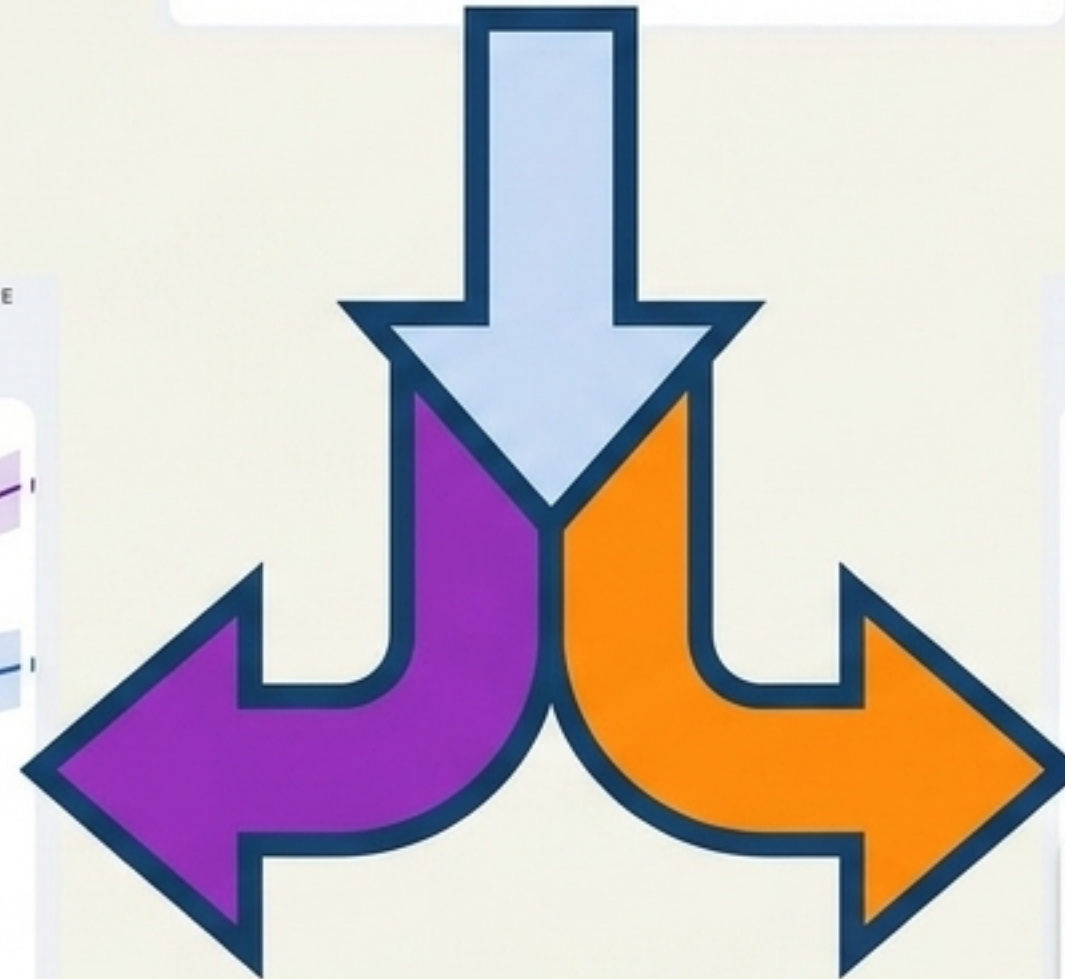
What it is:	Linear Model Plot
Level:	Figure-Level function (uses 'FacetGrid' magic).
How it works:	It uses 'regplot' under the hood, but adds the superpower to split your data.
Capabilities:	It calculates and draws completely separate regression lines so you can compare trends. You can also use 'col' and 'row' to spawn multiple charts. Notice we CANNOT use 'ax=' here because it is Figure-Level!



The 'Hue' Superpower Fractures the Math



A single linear regression assumes everyone follows the same pattern.



By adding just one word (`hue`), Seaborn calculates distinct mathematical models. It reveals how Gender interacts with Study Hours, showing completely separate trends.

The Structural Differentiation Matrix

Feature	<code>sns.regplot()</code>	<code>sns.lmplot()</code>
Full Name	Regression Plot	Linear Model Plot
Scope	Unified trend (one line)	Split trends (multiple lines via hue)
Hierarchy	Axes-Level	Figure-Level
Under the Hood	The basic building block	Uses <code>regplot</code> + <code>FacetGrid</code> magic
Embedding	CAN use <code>ax=</code> parameter	CANNOT use <code>ax=</code> parameter
Sub-plot Spawning	Cannot use <code>col</code> or <code>row</code>	Uses <code>col</code> and <code>row</code> to spawn multiple charts

Rule of Thumb Decision Tree

What is your goal for this visualization?

I just need a quick trendline on a single chart.

Use `sns.regplot()`

I want to compare trends between different categories or spawn multiple charts.

Use `sns.lmplot()`
Leverage `hue`, `col`, or `row`

Remember: They calculate the exact same math, you are just choosing the right structural tool for the job.